

2.3 Fields that derive from a potential

23/09/2026

2.3.1 Description of conservative fields

• Definition: let $\Omega \subset \mathbb{R}^n$ be an open domain, and

$$F: \Omega \rightarrow \mathbb{R}^n$$

$$x \mapsto F(x) = (F_1(x), \dots, F_n(x))$$

a vector field, $F \in C^1(\Omega, \mathbb{R}^n)$.

We say that F derives from a potential in Ω if $\exists$ a scalar field

$$f: \Omega \rightarrow \mathbb{R} \quad \text{of class } f \in C^1(\Omega) \text{ s.t.}$$
$$x \mapsto f(x)$$

$$F = \text{grad}(f) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$$

In this case, F is a conservative field, and f is a potential.

Remarks:

1) Example of the gravitational field

$$f = \frac{c}{r}, \quad c = g m M, \quad F = \text{grad } f = -\frac{c}{r^3} (x, y, z)$$

2) If the potential exists, it is defined up to a constant $\alpha \in \mathbb{R}$.

$$F = \text{grad}(f), \quad F = \text{grad}(f + \alpha)$$

2.3.2 Important results

• Theorem 1: let $\Omega \subset \mathbb{R}^n$ be an open domain and

$$F: \Omega \rightarrow \mathbb{R}^n \\ x \mapsto F(x) \quad \text{a vector field s.t. } F \in C^1(\Omega, \mathbb{R}^n)$$

a) Necessary condition: If F derives from a potential in Ω

$$\text{then } \frac{\partial F_i}{\partial x_j}(x) = \frac{\partial F_j}{\partial x_i}(x) \quad \forall i, j = 1, \dots, n \quad (*)$$

and $\forall x \in \Omega$

b) Sufficient condition: If $(*)$ holds and if Ω is simply connected and/or convex then F derives from a potential in Ω .

Remarks:

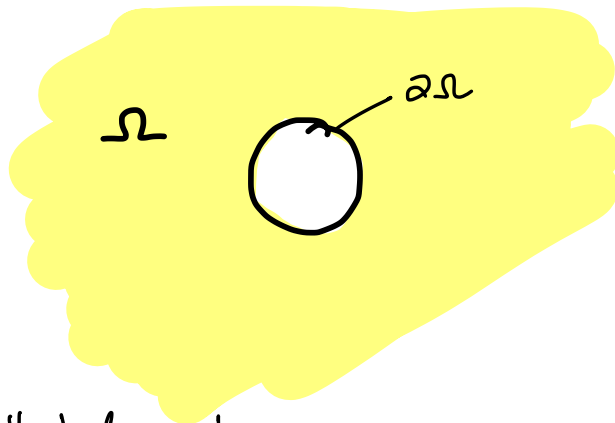
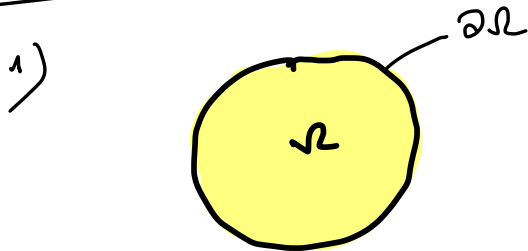
1) The condition (*) is necessary but not sufficient.

2) The condition (*) is equivalent to $\text{curl}(F) = 0$.

E.g. $n=2$ $\text{curl } F = \frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2} = 0$

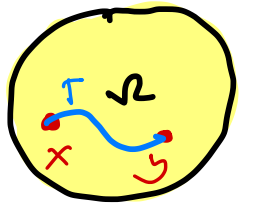
$\text{curl}(\underbrace{\text{grad } f}_F) = 0$

Recalls (or definitions):

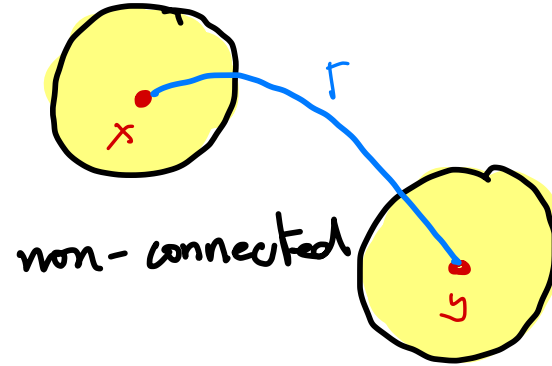


Ω is open if $\partial\Omega$ doesn't belong to Ω

2) $\Omega \subset \mathbb{R}^n$ is **connected** if $\forall x, y \in \Omega \exists$ a curve Γ that connects x and y and that is fully contained in Ω .



connected

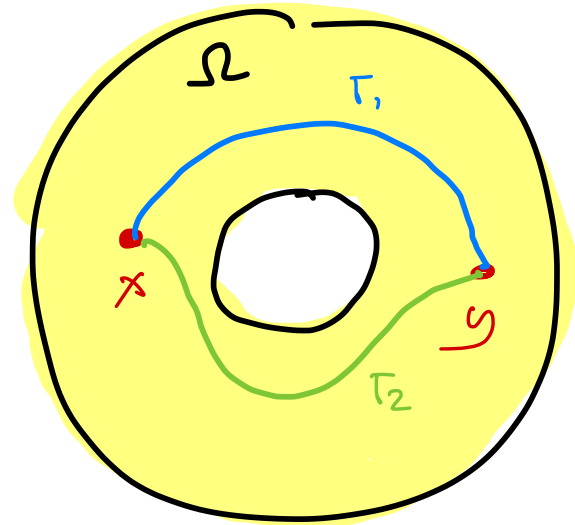
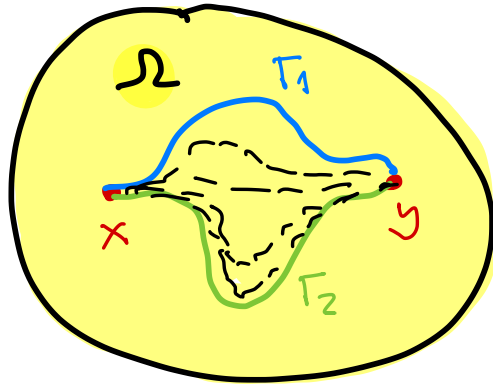


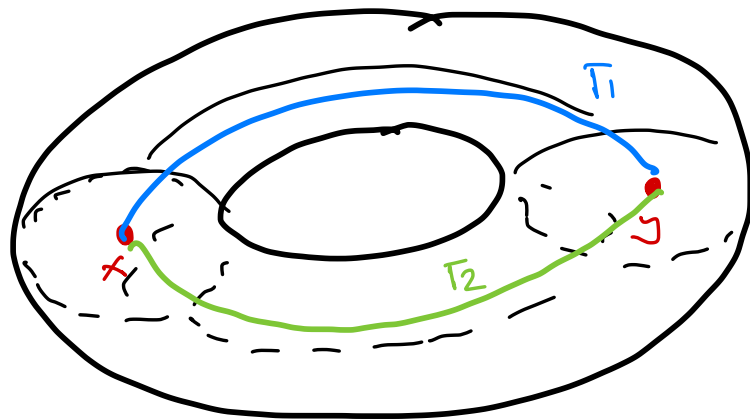
non-connected

3) $\Omega \subset \mathbb{R}^n$ is **simply connected** if $\forall x, y \in \Omega$ and

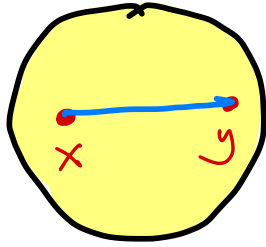
$\forall$ two curves $\Gamma_1, \Gamma_2 \subset \Omega$ that connect x and y ,

If I can deform one curve into the other such that along this process both curves stay inside Ω , then Ω is simply connected.

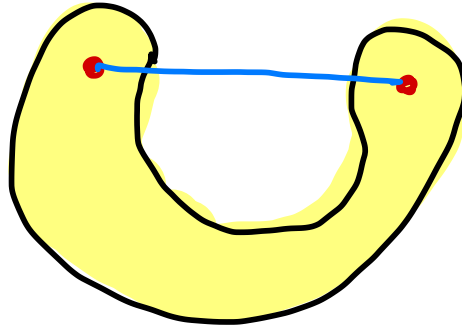




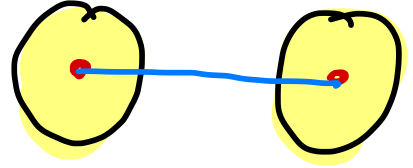
4) $\Omega \subset \mathbb{R}^n$ is **convex** if $\forall t \in [0, 1]$ and $x, y \in \Omega$
we have $x + t(y-x) \in \Omega$, i.e., the segment (straight)
between x and y is fully contained in Ω .



convex



non-convex



convex $\Rightarrow$ simply
connected.

• Theorem 2: Let $\Omega \subset \mathbb{R}^n$ be an open domain
and let $F: \Omega \rightarrow \mathbb{R}^n$ s.t. $F \in C^1(\Omega, \mathbb{R}^n)$

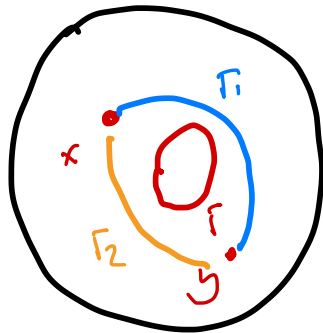
Then, the following statements are equivalent:

1) F derives from a potential

2) $\int_{\Gamma_1} F \cdot dl = \int_{\Gamma_2} F \cdot dl$ for all the (piecewise)

regular simple curves Γ_1 and $\Gamma_2 \subset \Omega$ joining every
couple of points x and $y \in \Omega$

3) $\int_{\Gamma} F \cdot dl = 0$ for every (piecewise) regular simple
curve Γ that is closed and $\Gamma \subset \Omega$.



Summary of thm 1 and 2: let be $\Omega \subset \mathbb{R}^n$ an open domain

let $F: \Omega \rightarrow \mathbb{R}^n$, s.t. $F \in C^1(\Omega, \mathbb{R}^n)$

Theorem 1

a) If $\text{curl } F \neq 0$ in $\Omega \Rightarrow F$ does not derive from a potential

b) If $\text{curl } F = 0$ in Ω

b1) if Ω is simply connected and/or convex

$\Rightarrow F$ derives from a potential

b2) otherwise (Ω is non simply connected, non convex)

$\Rightarrow$ Theorem 1 says nothing.

c) If we find at least one closed curve $T \subset \Omega$ s.t.
piecewise regular simple

$\int_T F \cdot dl \neq 0 \Rightarrow F$ does not derive from a potential

d) If we find one (or more) p.r.c. closed curves $T \subset \Omega$
s.t. $\int_T F \cdot dl = 0 \Rightarrow$ Theorem 2 says nothing.

Theorem 2

2.3. 2 Examples

• Example 1:

a) $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$(x, y) \mapsto F(x, y) = (4x^3y^2, 2x^4y + y)$$

- Does F derive from a potential?
- If so, find f s.t. $\text{grad } f = F$.

$\Omega = \mathbb{R}^2$, simply connected and convex. ✓

$$\text{curl } F = \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = 0 ?$$

$$\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x} = 8x^3y \quad \checkmark$$

in b1) $\rightarrow$ derives from a potential (theorem 1)

$$\text{grad } f = F = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (F_1, F_2)$$

$$1) \frac{\partial f}{\partial x} = F_1 = 4x^3y^2 \rightarrow \int \frac{\partial f}{\partial x} dx = \int 4x^3y^2 dx$$

$$\rightarrow f(x, y) = x^4y^2 + \alpha(y)$$

$$2) \frac{\partial f}{\partial y} = F_2 = 2x^4y + y$$

$$\frac{\partial}{\partial y} (x^4y^2 + \alpha(y)) = \cancel{2x^4y} + \alpha'(y) = F_2 = \cancel{2x^4y} + y$$

$$\alpha'(y) = y \rightarrow \int \alpha'(y) dy = \int y dy \rightarrow$$

$$\alpha(y) = \frac{1}{2}y^2 + C$$

$$f(x, y) = x^4 y^2 + \frac{1}{2} y^2 + c, \quad c \in \mathbb{R}$$

Example 2:

$$F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$(x, y, z) \mapsto F(x, y, z) = (y e^x \sin z, 1 + e^x \sin z, y e^x \cos z + z).$$

$\Omega = \mathbb{R}^3$. convex and simply connected ✓

$$\text{curl } F = 0? \quad \text{curl } F = \begin{pmatrix} \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \\ \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \\ \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \checkmark$$

b1) (Theorem 1) $\rightarrow$ derives from pot.

$$F = \text{grad } f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = (F_1, F_2, F_3)$$

$$1) \frac{\partial f}{\partial x} = F_1 = ye^x \sin z \rightarrow \int \frac{\partial f}{\partial x} dx = \int ye^x \sin z dx$$

$$\rightarrow f(x, y, z) = ye^x \sin z + \alpha(y, z)$$

$$2) \frac{\partial f}{\partial y} = F_2 = 1 + \cancel{e^x \sin z} = \frac{\partial}{\partial y} (ye^x \sin z + \alpha(y, z))$$

$$= \cancel{e^x \sin z} + \frac{\partial \alpha}{\partial y}, \quad \frac{\partial \alpha}{\partial y} = 1 \rightarrow \alpha(y, z) = y + \beta(z)$$

$$3) \frac{\partial f}{\partial z} = F_3 = \dots = \frac{\partial}{\partial z} (ye^x \sin z + \underbrace{y + \beta(z)}_{\alpha(y, z)})$$